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Predicting divorce using machine learning

A critical examination of variable Informativeness

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Abstract

Divorce represents one of the most significant family transitions in contemporary societies, with substantial consequences for the well-being of adults and children alike. The growing availability of relational and demographic data, combined with advances in machine learning, has prompted researchers to explore predictive modelling of marital dissolution. However, a fundamental question remains underexplored: what types of variables are necessary for such models to produce reliable predictions? This project critically examines the informational requirements for machine learning–based divorce prediction.

Using a dataset of 5,000 couples sourced from Kaggle, three classification algorithms (Logistic Regression, Random Forest, and XGBoost) were trained and evaluated on 21 features spanning demographic, socioeconomic, and self-reported relational characteristics. All three models converged near the random prediction baseline, achieving AUC values between 0.57 and 0.59 and F1-scores below 0.33 for the divorced class. The consistency of poor performance across architecturally distinct models indicates that the failure is attributable to the structure and signal content of the data rather than to any modelling limitation. Feature importance analysis further revealed no stable dominant predictor: variable rankings diverged substantially across models, with communication score, mental health issues, and cultural background match each ranking highly in some models but not others.

To address the gap between available data and the informational requirements of reliable prediction, a systematic synthesis of peer-reviewed evidence was conducted across nine thematic domains. The review identified behavioural interaction markers, particularly Gottman's Four Horsemen (contempt, criticism, defensiveness, stonewalling), longitudinal trajectories of relationship satisfaction, adult attachment styles, physiological stress responses during conflict, intergenerational transmission of divorce, and validated psychometric instruments as carrying the most consistently replicated predictive signal. These variable types are systematically absent from standard cross-sectional datasets, explaining the performance ceiling observed in this and similar studies.

The findings reframe the core challenge of divorce prediction as a data problem rather than an algorithmic one. Practical progress requires longitudinal, dyadic, and psychometrically grounded data collection strategies.

Keywords: divorce prediction, machine learning, feature importance, variable informativeness, relationship quality

1. Introduction

Context

Divorce has evolved from a rare and stigmatized event to one of the most common family transitions in contemporary societies. Over the second half of the twentieth century, marital dissolution increased dramatically across industrialized nations: the global divorce rate more than doubled between 1970 and 2008, rising from 2.6 to 5.5 divorces per 1,000 married persons (Fareed et al. 2022). This shift reflects big structural and normative changes in how societies conceive of marriage, individual autonomy, and gender equality. Drawing on panel data covering 84 countries from 1970 to 2008, Wang et al. (2018) demonstrated that exposure to global cultural institutions promoting individualism and human rights is strongly associated with higher divorce rates, even after controlling for economic development and legal frameworks (Wang et al. 2018).

Within Europe, divorce rates vary considerably across countries, reflecting differences in legal regimes, religiosity, welfare state provision, and marriage norms. A meta-analysis of over 260 European longitudinal studies by Wagner and Weiss (2006) confirmed substantial heterogeneity both within and between countries. Countries with weaker marriage norms, including the Nordic states and parts of Western Europe, tend to show weaker associations between standard risk factors and divorce, partly because the overall propensity to dissolve unions is higher and less socially constrained.

Switzerland occupies a particular position in this European landscape. According to the Swiss Federal Statistical Office (FSO/OFS), approximately 40% of Swiss marriages end in divorce, and the country has maintained a crude divorce rate of around 1.9 per 1,000 inhabitants in recent years, above the European Union average of approximately 1.8 (Swiss Federal Statistical and Eurostat 2022). The Swiss divorce rate rose following the liberalization of divorce law in 2000 (allowing unilateral divorce after a four-year separation) and peaked around 2010, before gradually stabilizing. Härkönen and Dronkers (2006), in a 17-country comparative analysis, found no significant educational gradient in divorce risk in Switzerland, suggesting that marital instability is more evenly distributed across social strata in Switzerland than in many other countries.

The social and economic costs of marital dissolution in Switzerland, as elsewhere, are substantial. Divorce is one of the leading causes of single-parent households and child poverty. (Amato 2000), In a landmark synthesis of the empirical literature, researchers concluded that marital dissolution has the potential to create considerable turmoil in the lives of both adults and children, with effects ranging from temporary well-being decrements to lasting downward socioeconomic trajectories. Leopold (2018) demonstrated significant gender asymmetries: while men tend to suffer greater short-term psychological distress following divorce, women bear disproportionate long-term economic consequences, including persistent income loss and heightened poverty risk. For children, the consequences are equally well-documented: in the United States, Seltzer (1994) found that parental divorce is associated with an average 37% decline in household income and significant disruption to social and emotional development, patterns that are broadly replicated in European contexts. These outcomes underscore the relevance of understanding and, potentially, anticipating marital dissolution.

Problem, research question and objectives

The growing availability of administrative and survey data, combined with advances in machine learning, has prompted researchers to explore predictive modelling of divorce. Several studies have demonstrated that, when data of sufficient quality is available, machine learning algorithms can achieve remarkably high prediction accuracy. Moumen et al. (2024) applied Random Forest, Naive Bayes, and artificial neural networks to a dataset structured around the Gottman Divorce Predictor Scale (DPS), achieving up to 91.66% accuracy. Fareed et al. (2022) similarly reported near-perfect performance (AUC close to 0.97) using ensemble methods on a question-based dataset designed by relationship specialists. Sharma et al. (2021)reached 98.5% accuracy using a Perceptron classifier on a Gottman-method dataset.

These high-performing models share a critical feature: they are trained on data capturing relational and psychological constructs (conflict patterns, communication quality, contempt, defensiveness) rather than on demographic and socioeconomic proxies alone. This distinction is not incidental. Heyman and Slep (2001) issued an early methodological warning that divorce prediction accuracy drops precipitously when models are cross-validated on new samples and when the base rate of divorce in the target population is considered. Their findings highlight that impressive in-sample accuracy can be highly misleading, and that the true test of a predictive model lies in its ability to generalize beyond the data on which it was trained.

Despite this growing body of work, a fundamental question remains underexplored: what types of data and variables are necessary for machine learning models to produce reliable divorce predictions? This question shifts the original research interest from "which model best predicts divorce?" to "what data would be needed to make divorce prediction reliable?" This question is not merely methodological, it has direct implications for data collection strategies, ethical considerations in predictive analytics, and the practical deployment of early warning tools in counselling and social policy contexts.

This project aims to critically examine the informational requirements for machine learning–based divorce prediction by evaluating which categories of variables carry sufficient predictive signal, and identifying the data gaps that limit model performance when only cross-sectional, population-level features are available.

Specifically, the project pursues three objectives:

- Evaluate the limitations of standard multidimensional features (demographic, socioeconomic, and self-reported relational metrics) in accurately predicting marital dissolution.
- Identify which feature domains contribute the most consistent signal across models, using cross-model feature importance analysis.
- Propose variable types that the divorce prediction literature identifies as truly discriminative.

2. Data

2.1. Data Provenance and Structure

The dataset used in this study was obtained from Kaggle, a widely used online platform that provides datasets for data science, machine learning, and research purposes. The dataset focuses on divorce prediction and contains information collected from 5000 couples. It consists of 5 features and 22 variables that describe different aspects of relationships and marriage status. The variables are designed to capture patterns and behaviours that may contribute to marriage status or divorce outcomes.

This dataset is synthetic and not based on real individuals. It is inspired by empirical research findings and intended for educational, experimental, and research use only.

The target variable in the dataset is Divorce, which is represented as a binary variable. The values are classified into two categories: divorce and non-divorce cases. This binary structure makes the dataset suitable for classification tasks in machine learning, where predictive models can be developed to determine the likelihood of divorce based on the provided variables. The remaining variables are features that help predict divorce outcomes, and they help explain relationships, behaviours, and communication patterns between couples.

2.2. Data Flow

The data flow process began with downloading the dataset from Kaggle and importing it into a Visual Studio environment for further exploration and preprocessing. After loading the dataset, an initial inspection was conducted to understand the dataset structure, variable types, dimensions, and overall data quality. Descriptive statistics and visualizations were used to examine the distributions of variables and identify patterns from the results.

The dataset was prepared for analysis through preprocessing and cleaning procedures. Since the dataset was already cleaned and well-structured, no major corrections were required. The dataset did not contain any missing values, duplicated records, or inconsistent data entries, which simplified the preprocessing stage.

2.3. Data Quality

In this dataset, the overall data quality was considered normal. The dataset didn’t have any missing values, which reduced the need for imputation methods or removal of incomplete records. Additionally, there were no duplicated entries detected, ensuring that each observation represented a unique couple.

The variables were also consistently formatted, allowing smooth integration into statistical analysis and machine learning workflows. Since the data is synthetic, there were fewer issues related to human error, inconsistent survey responses, or incomplete information that are commonly found in real-world datasets.

2.4. Feature Engineering and Preprocessing

Preprocessing primarily focused on preparing the variables and splitting them into Numerical, categorical, and multi-categorical variables. Numerical variables were examined to ensure that they were within expected ranges and suitable for analysis. Categorical and multi-categorical variables were examined to ensure that the categories were properly encoded and ready for Machine Learning.

Exploratory Data Analysis (EDA) techniques, such as histograms and distribution plots, were used to better understand the variables characteristics. These visualisations helped identify patterns, trends, and relationships within the data. The Divorce variable was analysed to evaluate the balance between classes and understand the distribution of divorce outcomes.

2.5. Data Cleaning

Minimal cleaning was required because the dataset was already preprocessed before publication on Kaggle. There were no missing values or duplicate records in the dataset, which made the cleaning process easy. Nevertheless, the dataset was still reviewed carefully to verify consistency and ensure that all variables were correctly interpreted. Variable names, data types, and category labels were checked to confirm accuracy and compatibility with analysis tools.

3. Exploratory data analysis

3.1. Descriptive statistics (Numerical variables)

The summary statistics of the sample(Table 1) indicate that participants married at an average age of 27.58 years, with most marrying between 24 and 31 years. Average marriage duration was 9.17 years, although variation was high, ranging from 1 to 40 years. Respondents had an average of 1.55 children, with most having one or two. Combined income averaged 60,220, with a wide range from 10,00 to 126,90. Communication, social support, and trust scores were relatively similar, averaging around 6 out of 10, suggesting moderate relationship quality. Financial stress was also moderate, averaging 5.07. Conflict frequency was generally low, with a median of 2. Shared hobbies averaged about three.

Table 1. Summary statistics

Variable	Mean	Std	Min	Q1	Median	Q3	Max
Age at marriage	27.58	4.86	18	24	28	31	45
Marriage duration (yrs)	9.17	8.85	1	2	6	13	40
Number of children	1.55	1.25	0	1	1	2	6
Combined income	60,220	19,66	10,00	46,66	60,19	73,67	126,90
Communication score	6.04	1.96	1.0	4.66	6.06	7.43	10.0
Conflict frequency	2.01	1.41	0	1	2	3	9
Financial stress level	5.07	2.33	1.0	3.37	5.06	6.72	10.0
Social support	5.97	1.97	1.0	4.60	6.01	7.33	10.0
Shared hobbies count	2.97	1.71	0	2	3	4	10
Trust score	6.03	1.93	1.0	4.70	6.03	7.37	10.0

The numerical variables show different distributional patterns (Figure 1). Age at marriage is approximately bell-shaped, centered around 28 years, with most observations between the mid-20s and early 30s. Marriage duration is strongly right-skewed: many marriages are relatively recent, while a small number extend up to 40 years. Number of children, shared hobbies count, and conflict frequency, concentrated at low values and positively skewed. Combined income appears roughly normally distributed around 60,000, with some lower and higher outliers. Communication score, social support, and trust score are broadly centered and show near-symmetric distributions. Financial stress is more widely spread.

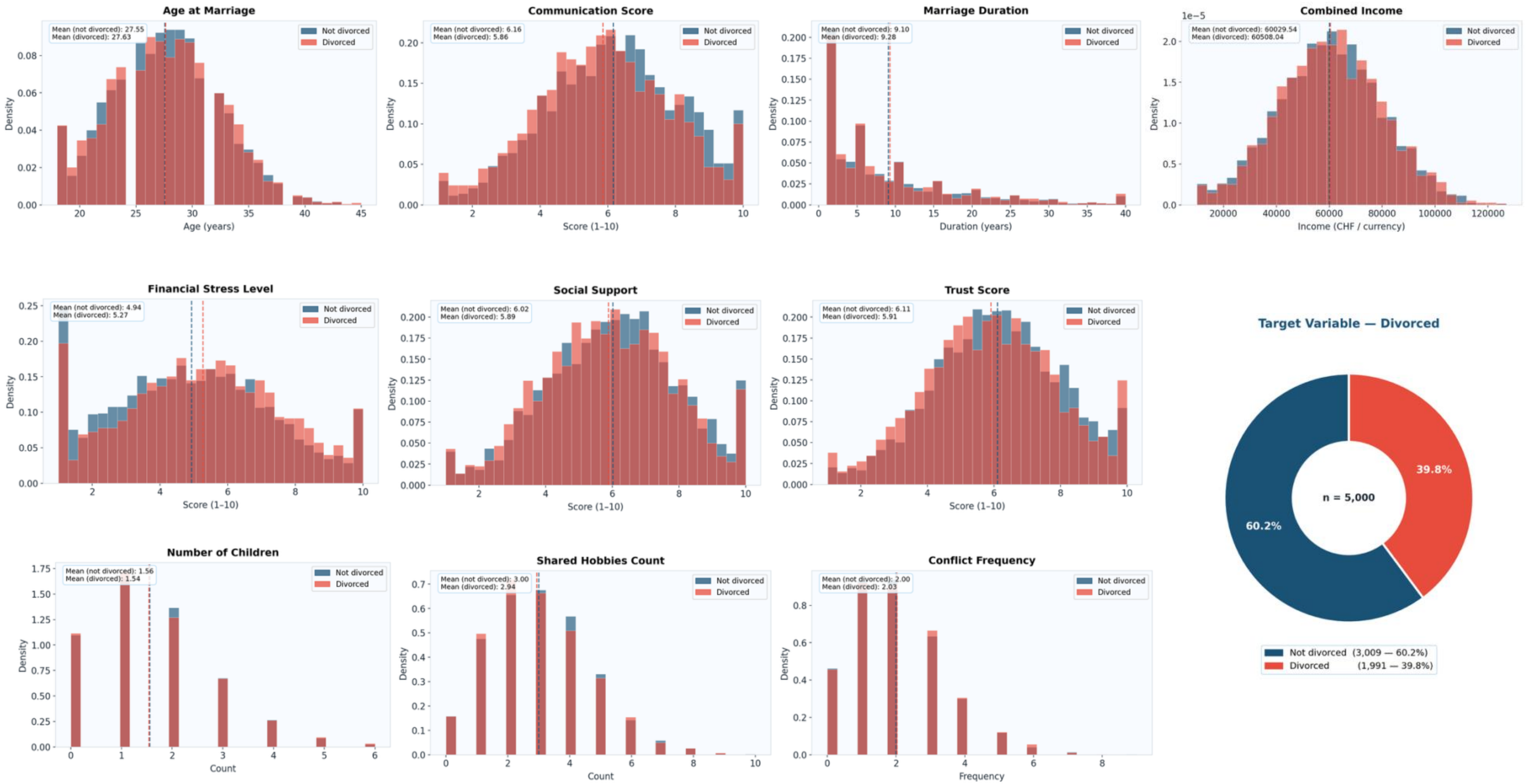

Figure 1. Data overview and distribution of numerical varibles

The boxplots (Figure 2) show substantial overlap between divorced and non-divorced individuals across most variables. Age at marriage, number of children, combined income, and shared hobbies appear very similar between groups, suggesting limited differences in demographic and household characteristics. Marriage duration is slightly higher among divorced individuals, with several long-duration outliers in both groups. Relationship-related variables show clearer contrasts: divorced individuals tend to report slightly lower communication, social support, and trust scores. They also appear to have somewhat higher financial stress and conflict frequency. Overall, the figure suggests modest group differences, mainly related to relationship quality and stress rather than demographic characteristics.

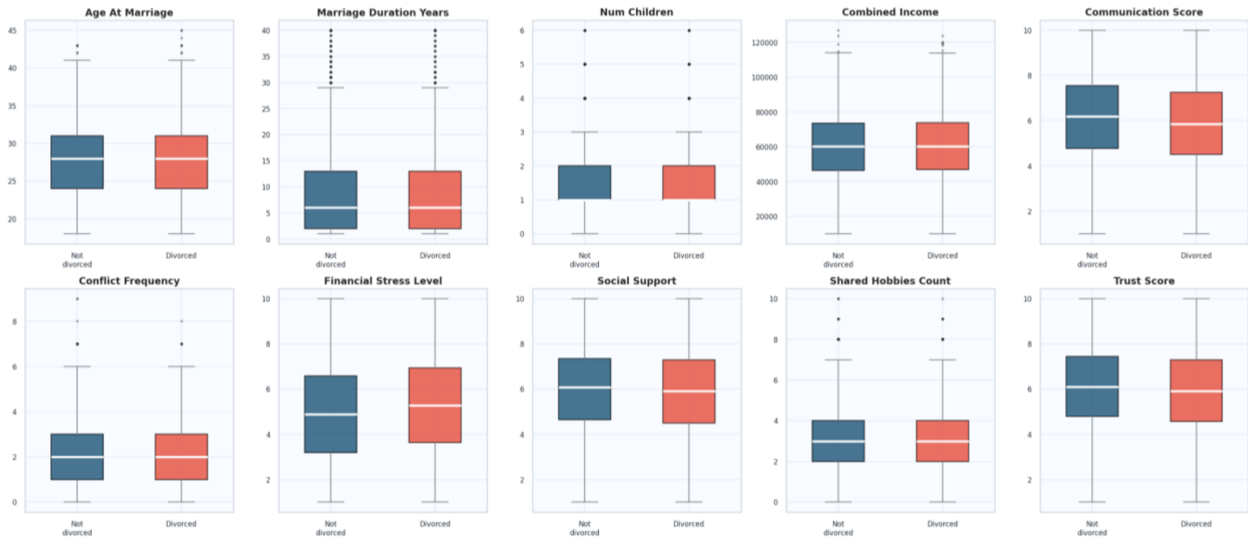

Figure 2. Boxplots of numerical variables by divorce status

The correlation matrix shows that the numerical variables are almost uncorrelated with each other. Most coefficients are very close to zero, generally ranging between about -0.04 and 0.04, indicating no meaningful linear relationships among age at marriage, marriage duration, number of children, income, communication, conflict, stress, social support, shared hobbies, and trust.

The target variable, divorced, also shows only very weak correlations with the numerical predictors. The strongest associations are still small: communication score is negatively correlated with divorce (-0.08), financial stress is positively correlated (0.07), and trust score is negatively correlated (-0.05). Overall, divorce is not strongly explained by any single numerical variable in this matrix.

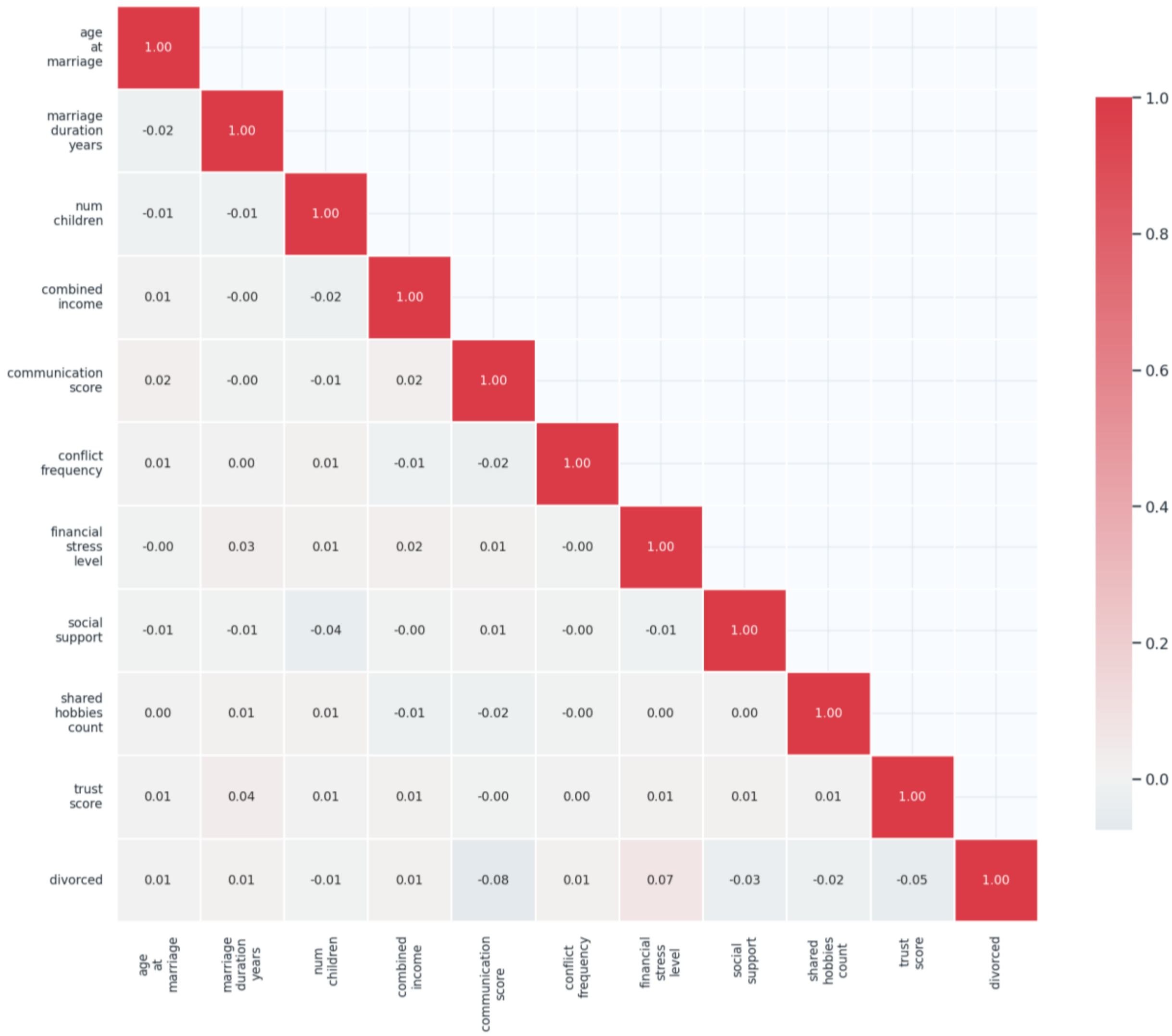

Figure 3. Correlation matrix

3.2. Descriptive statistics (Categorical and binary variables)

Regarding the categorical and binary variables, several of them show higher divorce rates in specific categories (Figure 4). Divorce appears more frequent among respondents reporting domestic violence history, infidelity occurrence, mental health issues, and possibly counseling attendance, suggesting these factors may be associated with marital instability. In contrast, premarital cohabitation and cultural background match show smaller differences in divorce rates. For education, employment, religion compatibility, marriage type, and conflict-resolution style, divorced and non-divorced respondents are present across all categories, with no single category fully distinguishing the groups.

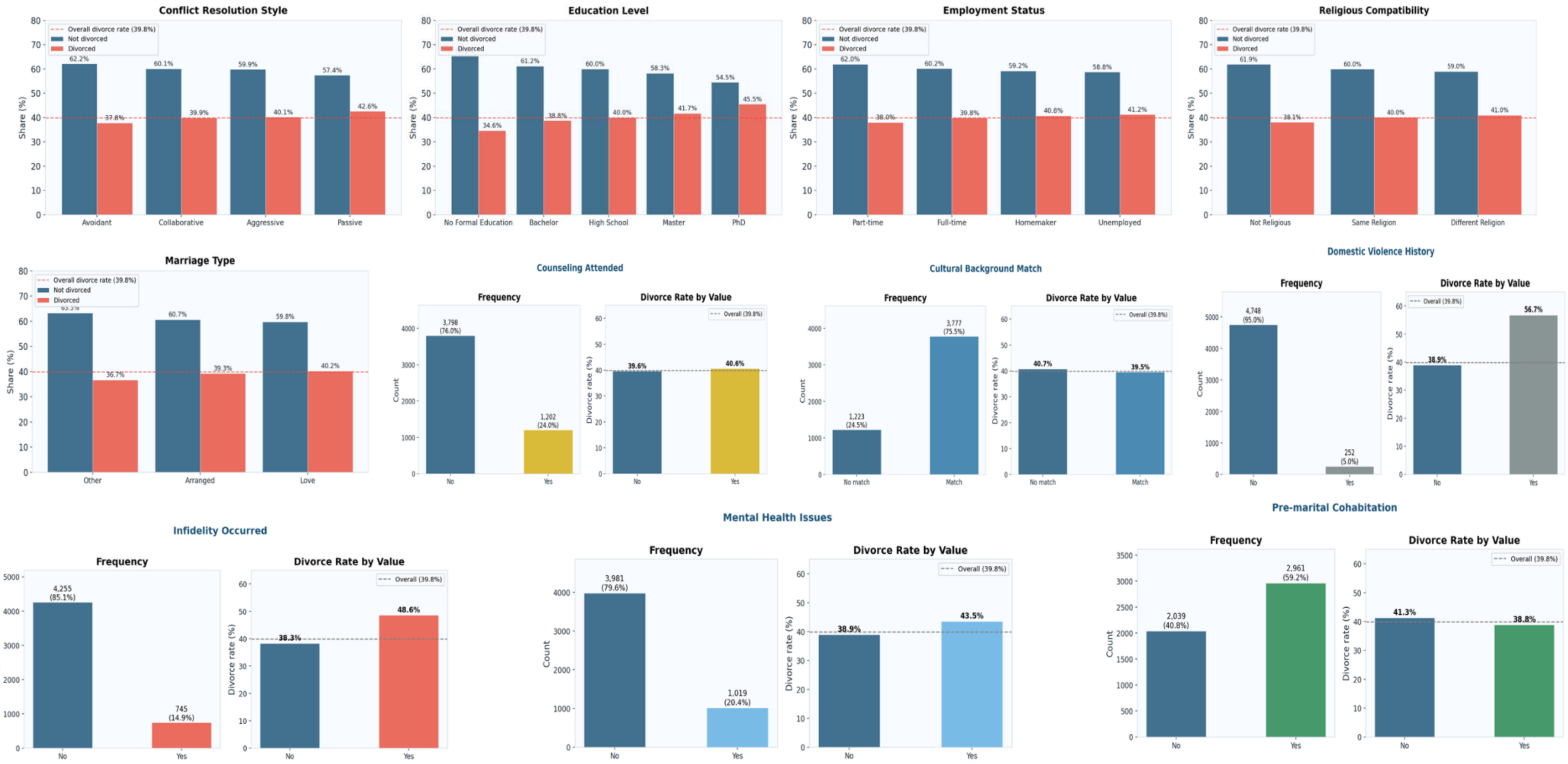

Figure 4. Categorical and binary variables value by divorce status

4. Machine learning

4.1. Setting up

The experimental setting up consisted of three steps:

- Data split. The dataset (5,000 observations, 21 features) was split into a training set (80%, n = 4,000) and a test set (20%, n = 1,000) using stratified random sampling to preserve the original 60/40 class distribution. The training set is about 4,000 observations (60.2% not divorced, 39.8% divorced), the test set about 1,000 observations (60.2% not divorced, 39.8% divorced) and the random seed about 42 (reproducibility)
- A unified preprocessing pipeline was applied consistently across the three machine-learning models: Logistic Regression, Random Forest, and XGBoost. The preprocessing was implemented using a "ColumnTransformer", allowing numerical and categorical variables to be processed separately within the same workflow. The ten numerical features were standardized using "StandardScaler", transforming them to have a mean of zero and a unit variance. The eleven categorical features were encoded using "OneHotEncoder", which converted each category into binary indicator variables. Each model was then embedded within a scikit-learn Pipeline, ensuring that all preprocessing steps were learned only from the training data and subsequently applied to the test data.
- Evaluation metrics. Given the moderate class imbalance (60/40), accuracy alone is misleading. We therefore also use Precision, Recall and F1-Score.

4.2. Model 1: Logistic regression

The Logistic Regression model achieved an overall accuracy of 61.2%, indicating moderate predictive performance. However, the model performed substantially better in identifying individuals who were not divorced than those who were divorced. For the "Not divorced" class, precision was 0.62 and recall was high at 0.89, meaning that most non-divorced cases were correctly detected. In contrast, performance for the "Divorced" class was weak, with a recall of only 0.19 and an F1-score of 0.28. This indicates that the model missed most divorced individuals, despite a moderate precision of 0.54. The macro-average F1-score of 0.51 confirms an imbalance in predictive performance across the two classes.

Table 2. Classification report for logistic regression

Class	Precision	Recall	F1-Score	Support
Not divorced	0.62	0.89	0.73	602
Divorced	0.54	0.19	0.28	398
Accuracy (overall)			0.612	1,000
Macro avg	0.58	0.54	0.51	

4.3. Model 2: Random Forest

The Random Forest model achieved an overall accuracy of 60.7%, indicating moderate but limited predictive performance. The model performed much better in identifying individuals who were not divorced than those who were divorced. For the "Not divorced" class, recall was high at 0.90, showing that most non-divorced individuals were correctly classified. However, the model performed poorly for the "Divorced" class, with a recall of only 0.16 and an F1-score of 0.25. This means that most divorced individuals were missed by the model. The macro-average F1-score of 0.49 confirms an important imbalance in performance between the two classes.

Table 2. Classification report for Random Forest

Class	Precision	Recall	F1-Score	Support
Not divorced	0.62	0.90	0.73	602
Divorced	0.52	0.16	0.25	398
Accuracy (overall)			0.607	1,000
Macro avg	0.57	0.53	0.49	

4.4. Model 3: XGBOOST

The XGBoost model achieved an overall accuracy of 58.7%, indicating limited predictive performance. The model classified “Not divorced” individuals better than “Divorced” individuals, with a precision of 0.62, recall of 0.81, and F1-score of 0.70 for the “Not divorced” class. However, its performance for the “Divorced” class remained weak, with a precision of 0.46, recall of 0.25, and F1-score of 0.32. This means that the model correctly identified only one quarter of divorced individuals. Although XGBoost improved recall for the divorced class compared with the other models, its overall accuracy was lower, and the macro-average F1-score of 0.51 indicates modest and imbalanced performance.

Table 3. Classification report for Random Forest

Class	Precision	Recall	F1-Score	Support
Not divorced	0.62	0.81	0.70	602
Divorced	0.46	0.25	0.32	398
Accuracy (overall)			0.587	1,000
Macro avg	0.54	0.53	0.51	

4.5. Models comparison

The comparison of the three models indicates generally limited predictive performance, with all AUC values below 0.60. Logistic Regression achieved the highest accuracy (61.2%) and AUC (0.5873), making it the best overall model in terms of general discrimination. Random Forest performed similarly in accuracy (60.7%) but showed the weakest recall for divorced cases (16.3%), missing most individuals in this class. XGBoost had the lowest accuracy (58.7%) and AUC (0.5746), but achieved the highest recall (24.9%) and F1-score (0.3241) for the divorced class. The confusion matrices confirm that all models were better at predicting “not divorced” cases than “divorced” cases. Overall, Logistic Regression offers the best overall balance, while XGBoost is slightly better at detecting divorced individuals.

All three models performed within the near-random prediction zone. Even the best-performing model, Logistic Regression, achieved an AUC only 0.087 above the random baseline of 0.50, which is insufficient for any practical predictive application. The fact that three fundamentally different model architectures converged toward nearly identical performance suggests that model architecture is not the main explanation for the poor results. Rather, the failure appears to be driven by the structure and quality of the dataset itself. The features were synthetically generated without a genuine data-generating process linking them meaningfully to the outcome. Therefore, the weak model performance should not be interpreted as a tuning problem, but as a data quality and signal problem. No amount of hyperparameter optimisation or additional model complexity can recover meaningful predictive performance from a dataset that lacks genuine predictive structure.

5. Feature importance

To answer the second research question, feature importance analysis was performed using the Logistic Regression, Random Forest, and XGBoost models. The objective was to identify the variables that contribute most to divorce prediction. The feature-importance results show that no single predictor clearly dominates the classification of divorce status.

For Logistic Regression, the coefficients were generally close to zero (Figure 5), confirming weak linear effects. Cultural background match, combined income, education level, premarital cohabitation, employment status, number of children, domestic violence history, and financial stress level had positive coefficients, indicating an association with higher predicted probability of divorce. In contrast, trust score, communication score, marriage duration, marriage type, conflict frequency, and social support had negative coefficients, suggesting an association with lower predicted divorce probability. The strongest negative coefficients were observed for communication score and trust score.

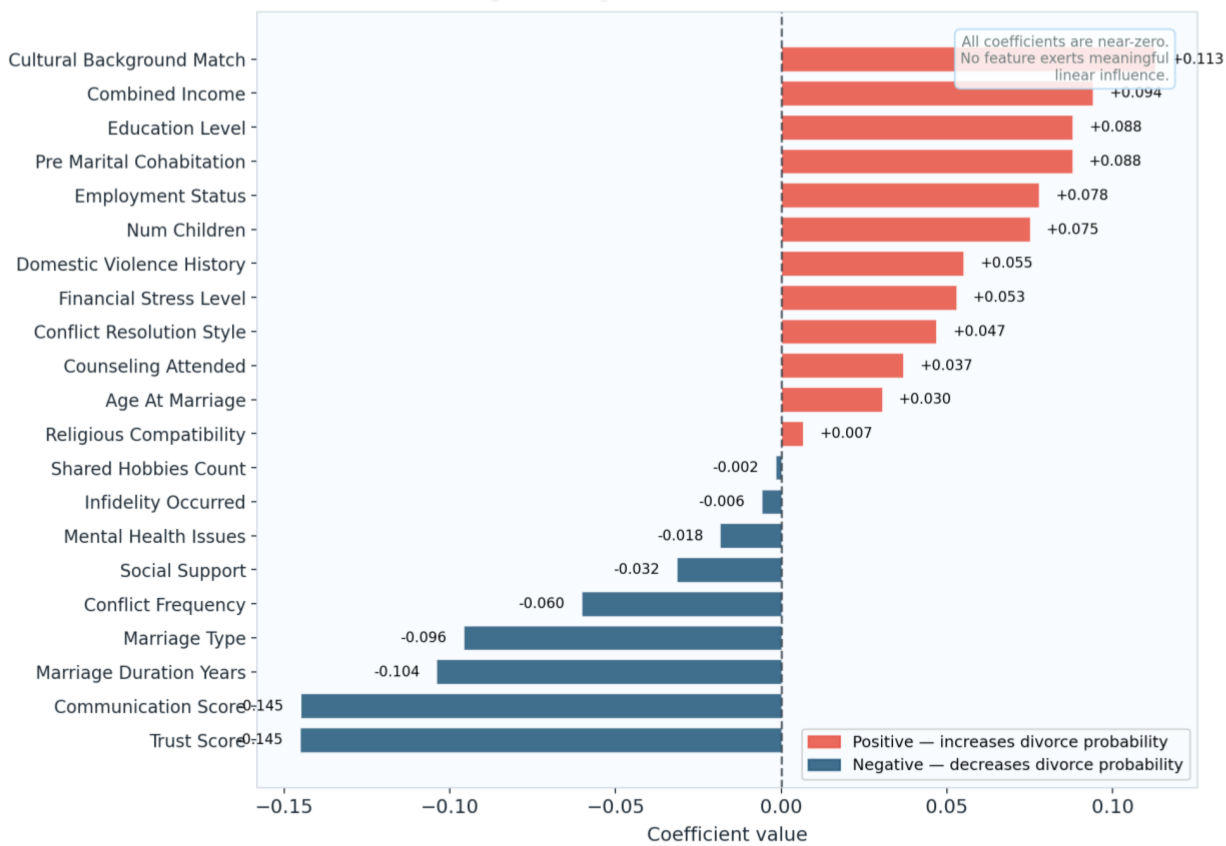

Figure 5. Feature importance coefficients for Logistic Regression

In the Random Forest model (Figure 6), communication score was the most important feature, followed by religious compatibility, mental health issues, conflict frequency, social support, age at marriage, employment status, and shared hobbies count. As with XGBoost, the differences between importance scores were relatively small, indicating weak feature dominance and a broadly distributed contribution of predictors.

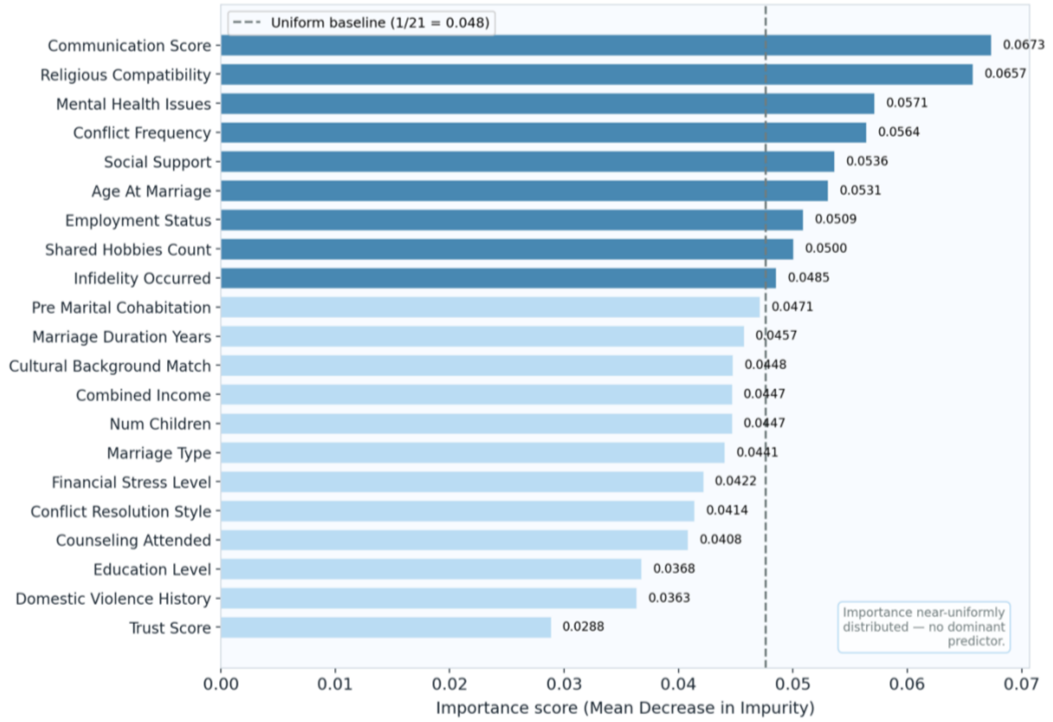

Figure 6. Feature importance coefficients for Random Forest

In the XGBoost model (Figure 7, the most important variables based on gain were mental health issues, shared hobbies count, cultural background match, conflict resolution style, domestic violence history, and combined income. However, the importance scores were close to the uniform baseline, suggesting that XGBoost distributed predictive information across many variables rather than relying on one strong predictor.

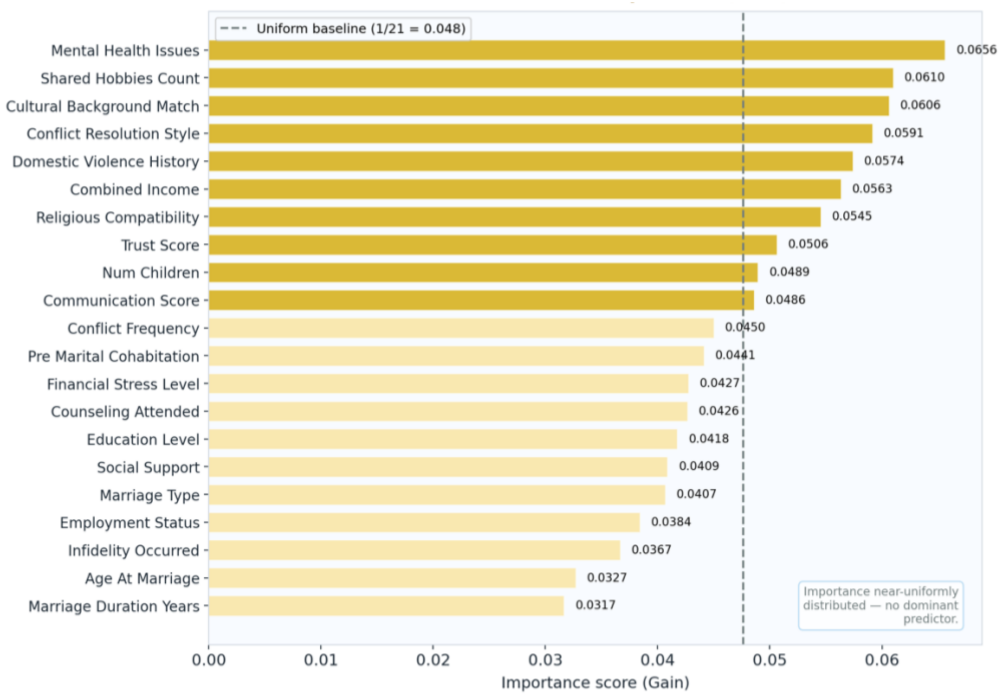

Figure 7. Feature importance coefficients for XGBOOST

The cross-model comparison (Figure 8) shows that feature rankings were not fully consistent across models. Communication score ranked highly in Logistic Regression and Random Forest but lower in XGBoost. Mental health issues ranked highly in Random Forest and XGBoost but not in Logistic Regression. Cultural background match was important in Logistic Regression and XGBoost but less so in Random Forest. Trust score was the most important Logistic Regression predictor but ranked last in Random Forest. Overall, the comparison suggests that the models captured different patterns in the data, but the near-uniform importance distribution and weak coefficients indicate that divorce status is not strongly driven by a small set of predictors.

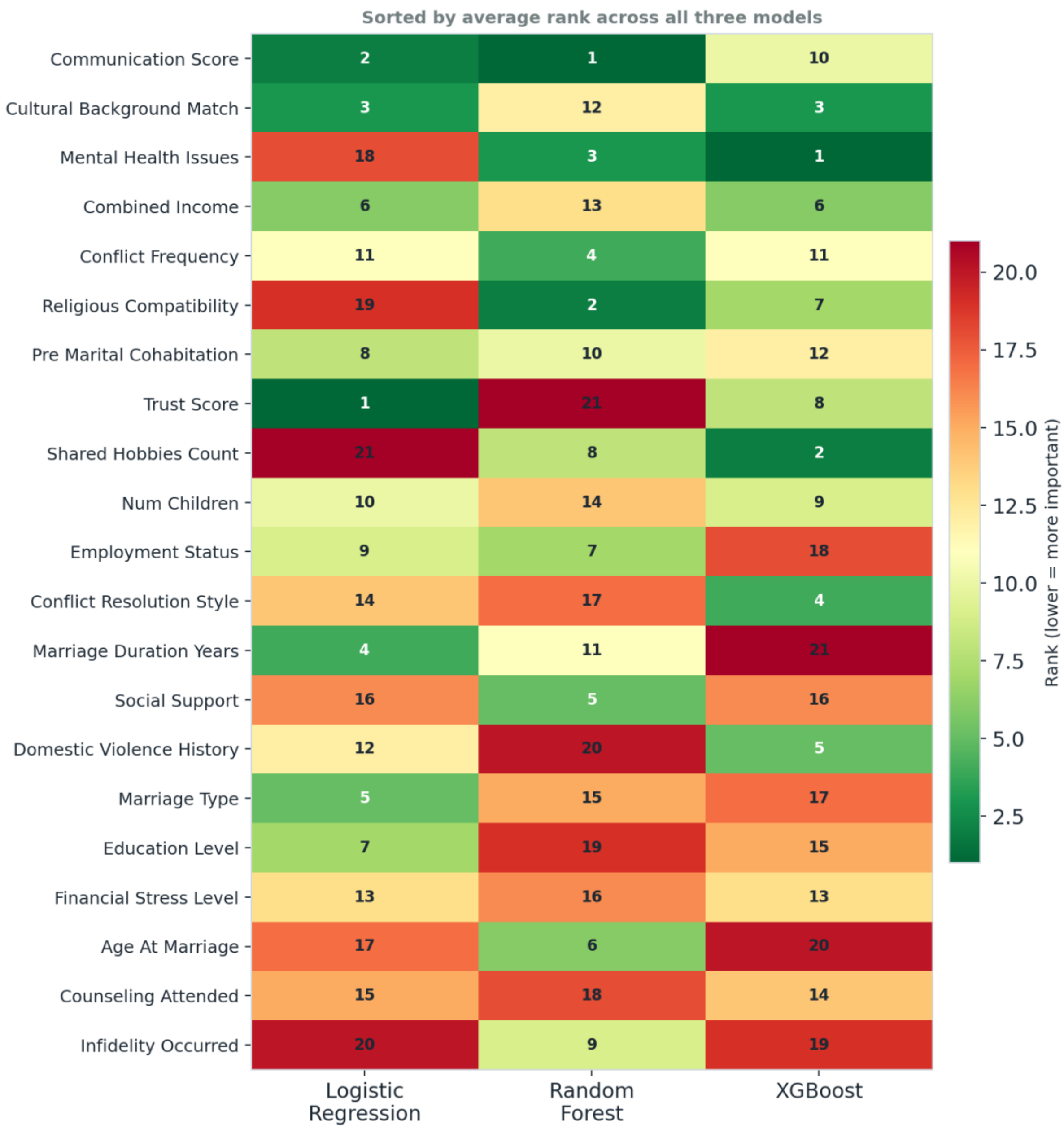

Figure 8. cross-model feature rank comparison (1 = most important)

6. Proposition of new discrimative variables

As for the third research objective, we conducted a systematic review synthesis of peer-reviewed evidence to identify additional variables with potential discriminative power for predicting marital dissolution beyond the standard cross-sectional demographic and socioeconomic features commonly used in machine-learning studies. The review covered nine thematic domains: behavioural interaction markers, longitudinal relationship satisfaction trajectories, physiological markers, attachment styles, intergenerational transmission of divorce, life events and stressors, sexual satisfaction and intimacy, communication behaviour, and validated psychometric instruments.

The strongest and most consistently replicated predictors of divorce relate to observed behavioural interaction patterns between partners. Gottman and Levenson's longitudinal study of 73 couples showed that marital dissolution could be predicted through a "cascade" of behavioural and physiological markers, including negative emotional expression, stubbornness, withdrawal, defensiveness, and reduced positive affect (Gottman and Levenson 1992). This framework was later formalised into the "Four Horsemen" of relationship dissolution: contempt, criticism, defensiveness, and stonewalling (Gottman 2023). Recent evidence confirms the central role of these behaviours. In a network analysis of 780 Iranian married couples, contempt emerged as the most central and influential variable in the marital conflict network (Hashemi et al. 2025). McNelis and Segrin (2019) further showed that dysfunctional communication patterns were strongly associated with insecure attachment and divorce history. However, these behavioural variables require structured couple interaction tasks and trained observer coding, meaning they are typically absent from conventional survey-based datasets.

A second high-value domain concerns longitudinal trajectories of relationship satisfaction. Rather than measuring satisfaction at a single point in time, the literature shows that the direction and speed of change are more informative. Birditt et al. (2012), following 373 couples over 16 years, identified distinct happiness trajectories and showed that declining or moderate-to-low trajectories significantly predicted divorce. Bühler et al. (2025), using four national longitudinal studies, found a consistent "terminal decline" pattern in which relationship satisfaction gradually decreases before dropping sharply between 0.58 and 2.30 years before separation. Similarly, van Scheppingen et al. (2019), using 33 waves of the German Socio-Economic Panel Study, found gradual pre-divorce declines followed by a sharp drop in the year of divorce. These findings suggest that repeated measures of relationship quality, ideally collected every six months to one year, are essential for estimating slope, curvature, and inflection points. Cross-sectional datasets cannot capture this signal.

Physiological markers during conflict represent another theoretically grounded predictor domain. Gottman and Levenson (1992) linked marital outcomes to physiological regulation, showing that couples on a dissolution trajectory exhibited distinctive cardiovascular responses during interaction. Coutinho et al. (2017) extended this evidence by measuring skin conductance, heart rate, and salivary cortisol during couple interactions, finding higher heart rate and cortisol during negative interactions and evidence of autonomic arousal under conflict. These findings indicate that physiological reactivity may provide objective markers of marital strain. However, such measures require laboratory protocols or wearable biosensors, limiting their immediate feasibility in standard population surveys.

Adult attachment style is a more feasible psychological predictor. Secure, anxious, avoidant, and disorganised attachment patterns shape communication, conflict resolution, intimacy, and relationship stability. McNelis and Segrin (2019) found that anxious and avoidant attachment predicted divorce history and current relationship status. Okonkwo (2024) showed that secure attachment was associated with better communication, more effective conflict resolution, and higher relationship satisfaction, while insecure attachment was linked to greater conflict. Davarinejad et al. (2022) also found that attachment anxiety and avoidance predicted poorer divorce adjustment. Since attachment style can be measured using validated self-report instruments such as the Experiences in Close Relationships scale or the Adult Attachment Questionnaire, its absence from many datasets is a correctable gap.

The intergenerational transmission of divorce is another robust predictor. Amato (1996) showed that parental divorce increases offspring divorce risk, particularly when the wife or both spouses experienced parental divorce, with the effect mediated mainly through interpersonal behaviour problems. More recent population-register studies support this association. Stanfors et al. (2024) found stable intergenerational transmission of divorce in Swedish register data spanning 1920–2015, with stronger transmission through the female line. Jørgensen et al. (2025), using adoption and children-of-twins designs in Norway, provided evidence for both environmental and genetic pathways. Parental divorce history is therefore a simple but high-signal binary variable that could be easily integrated into surveys.

Life events and stressors also operate as proximal triggers of divorce. Amato and Rogers (1997) found that marital problems such as infidelity, irresponsible spending, substance use, jealousy, moodiness, and irritating habits predicted divorce up to 12 years later. Keldenich and Knabe (2020) showed that husbands' involuntary job loss increased marital dissolution risk by 54–74%, even after re-employment, whereas wives' job loss did not have a comparable effect. Rezapour (2024) found that traumatic life events, including infidelity, abuse, and disability, were positively associated with divorce. These findings highlight the need for event-based longitudinal variables that capture both occurrence and timing.

Sexual satisfaction and intimacy also show predictive relevance. Yabiku and Gager (2009), using data from 5,902 respondents, found that low sexual frequency was associated with higher union dissolution, especially among cohabiting couples. Gheshlaghi et al. (2014) found that sexual satisfaction was significantly associated with divorce status among Iranian women applying for divorce. However, these measures may be affected by social desirability bias.

Communication behaviour is important but insufficient when measured only once. Kanter and Proulx's meta-analysis of 64 longitudinal dyadic studies found that negative communication had a statistically significant small-to-moderate effect on later dissolution, while positive communication had a smaller and non-significant effect (Kanter and Proulx 2021). This supports the idea that communication should be combined with attachment, physiological regulation, and longitudinal relationship-quality measures.

Finally, validated psychometric instruments offer a major opportunity to improve prediction. The Dyadic Adjustment Scale captures satisfaction, cohesion, consensus, and affectional expression and remains a gold standard for relationship quality measurement (Spanier 1976). Sabourin et al. (2005) showed that the abbreviated DAS-4 predicted couple dissolution as effectively as the full version. Other validated tools, such as the Kansas Marital Satisfaction Scale and ENRICH Couple Scales, similarly capture relational constructs more directly than demographic proxies.

The table below summarises the nine variable domains identified in this review, the specific variables within each domain, the assessed strength of evidence, and the data type required for their measurement.

Table 4. Discriminative domains and variables for divorce prediction

Domain	Key variables	Evidence strength	Data type required
Behavioural interaction	Contempt, criticism, defensiveness, stonewalling	**Very strong**	Coded video / audio interaction
Longitudinal satisfaction trajectories	Rate of decline, terminal phase onset	**Very strong**	Repeated measures (≥3 waves)

Physiological markers	Heart rate, skin conductance, cortisol during conflict	**Moderate – strong**	Lab biosensor / wearable data
Attachment style	Insecure (anxious / avoidant) vs secure	**Strong**	Validated scale (ECR, AAQ)
Intergenerational history	Parental divorce – especially wife's parents	**Very strong**	Life history record / register
Life events & stressors	Infidelity, job loss, substance abuse, trauma	**Strong**	Event-based longitudinal data
Sexual satisfaction	Frequency, satisfaction scores	**Moderate**	Self-report scales
Communication behaviour	Negative affect ratio, demand–withdraw patterns	**Moderate (d = −0.41)**	Coded interaction or validated scale
Psychometric dyadic scales	DAS, KMS, ENRICH scores	**Strong**	Validated dyadic questionnaire

7. Discussion

This project aimed to critically examine the informational requirements for machine learning–based divorce prediction by evaluating which categories of variables carry sufficient predictive signal, and identifying the data gaps that limit model performance when only cross-sectional, population-level features are available. The empirical results obtained from three supervised learning models (Logistic Regression, Random Forest, and XGBoost) are discussed below in direct relation to each of the three specific objectives.

7.1. Limits of standard multidimensional features in predicting divorce

The first objective was to evaluate the limitations of standard demographic, socioeconomic, and self-reported relational features in accurately predicting marital dissolution. The results provide clear and convergent evidence of these limitations. All three models achieved AUC scores between 0.575 and 0.587, barely above the 0.500 random baseline, and divorce-class recall ranging from only 0.16 to 0.25, meaning that between 75% and 84% of actual divorces were missed. These metrics confirm that the 21 features included in the dataset are insufficient for reliable classification, regardless of the model architecture applied.

The exploratory data analysis reinforces this diagnostic. Pearson correlations between all numeric features and the target variable were uniformly near-zero ($|r| < 0.05$), and distributional overlap between divorced and non-divorced groups was near-complete across every variable. These patterns are characteristic of datasets where the outcome has been generated independently of the predictors, a structural property consistent with the synthetic nature of the data (Iantovics et al. 2022). More broadly, however, they illustrate a known limitation of the variable types selected: demographic variables such as age at marriage, number of children, and combined income are distal risk factors that operate probabilistically at the population level but do not determine individual outcomes (Kanter and Proulx 2021). Self-reported relational scores such as communication_score and trust_score, while theoretically relevant, are measured at a single point in time and with unvalidated instruments, producing attenuation bias that suppresses predictive signal (Heyman and Slep 2001). The Dyadic Adjustment Scale, by contrast, has demonstrated criterion-related validity for couple dissolution in large longitudinal studies (Spanier 1976), underlining that measurement quality is a key determinant of model performance.

A further limitation lies in the cross-sectional design itself. Research consistently shows that the *trajectory* of relationship quality, rather than its level at any single point, carries the strongest predictive signal. Bühler et al. (2025) demonstrated a systematic terminal decline in relationship satisfaction in the 0.58 to 2.30 years immediately before separation (Bühler et al. 2025), a pattern that is structurally invisible to any cross-sectional dataset. The present results can therefore be understood as an empirical confirmation of what the design of the dataset makes impossible: reliable prediction from variables that lack temporal depth.

7.2. Contribution of intial features in predicting divorce

The second objective was to identify, through cross-model feature importance analysis, which feature domains contribute the most consistent signal across models. The results reveal that no domain clearly dominates, but some nuanced patterns deserve attention.

In the Logistic Regression model, the strongest negative coefficients — indicating association with lower predicted divorce probability, were observed for communication score and trust score. This directional finding is consistent with the meta-analytic evidence showing that negative communication behaviours have a significant, albeit modest, effect on dissolution (d = −0.41) (Kanter and Proulx 2021). Financial stress and infidelity occurrence showed the largest positive associations with divorce, in line with longitudinal evidence identifying these as proximal triggers (Amato and Rogers 1997). However, all coefficients remained near-zero in absolute magnitude, precluding substantive conclusions about effect sizes.

In Random Forest and XGBoost, importance scores were distributed near-uniformly across all 21 features, and the cross-model ranking comparison revealed substantial inconsistency: communication score ranked highly in Logistic Regression and Random Forest but lower in XGBoost, while mental health issues ranked highly in both tree-based models but not in Logistic Regression. This cross-model instability has been interpreted in the methodological literature as a diagnostic marker of weak and distributed signal, rather than genuine feature-specific effects (Fareed et al. 2022). It suggests that the models were responding to noise rather than to consistent patterns in the data.

A cautious reading of the cross-model evidence nevertheless points to *relational and psychological variables* (communication, trust, conflict frequency, mental health) as showing marginally greater consistency than purely demographic or socioeconomic features, which aligns with the theoretical expectation that proximal relational processes are more directly linked to dissolution than distal structural characteristics (Gottman and Levenson 1992). However, this pattern is too weak and inconsistent across models to constitute evidence of genuine domain-level signal in the present dataset.

7.3. New discrimative variables

The third objective was met through the systematic review of nine evidence domains conducted in Section 5 of this report. The review identifies a clear typology of variable types that the divorce prediction literature associates with strong discriminative power — all of which are absent from the current dataset.

The strongest and most consistently replicated predictors are observed behavioural interaction markers. Gottman and Levenson (1992) demonstrated that a cascade of behavioural and physiological signals, including contempt, criticism, defensiveness, and stonewalling, predicted marital dissolution in longitudinal follow-up (Gottman 2023). These "Four Horsemen" constructs are not measurable through survey proxies; they require structured couple interaction tasks and trained observer coding. A network analysis of 780

Iranian couples confirmed that contempt holds the highest centrality in the marital conflict network (Hashemi et al. 2025), further establishing its primacy as a predictive marker.

Beyond behavioural coding, the review identifies repeated measures of dyadic adjustment using validated instruments (e.g., the Dyadic Adjustment Scale (Spanier 1976)), attachment style assessed via the Experiences in Close Relationships scale [12]), parental divorce history (a simple binary variable with strong and stable intergenerational predictive validity (Amato 1996)), and event-based longitudinal records capturing the timing of critical stressors such as job loss, infidelity disclosure, and substance use onset (Amato and Rogers 1997) as the most feasible improvements over the current variable set. Of these, attachment style and parental divorce history are the most immediately actionable additions to survey-based datasets, as they require only brief self-report instruments rather than laboratory or register data.

In sum, the empirical results of this project, read against its three objectives, tell a coherent story: standard cross-sectional features fail to predict divorce because they lack the relational depth, temporal dynamics, and psychometric precision that the discriminative prediction task requires. The data requirements identified through the systematic review provide a concrete roadmap for constructing a dataset that would make reliable machine learning prediction of marital dissolution feasible (Wagner and Weiss 2006).

8. Conclusion

This project set out to investigate whether standard cross-sectional demographic and socioeconomic data are sufficient to build reliable machine learning models for divorce prediction, and to identify what kinds of variables would genuinely improve predictive performance.

Across three fundamentally different model architectures, Logistic Regression, Random Forest, and XGBoost, predictive performance converged near the random baseline. This near-uniform failure across model types demonstrates that the bottleneck is not algorithmic: no amount of tuning or model complexity could extract a signal that the data does not contain. The synthetic dataset, while well-structured and clean, lacked the genuine data-generating process that links predictors to outcomes in the real world. This renders model underperformance a feature of the data rather than a flaw of the approach.

Feature importance analysis reinforced this conclusion. No single predictor dominated across models, and the ranking of variables was inconsistent: communication score was the strongest predictor in Logistic Regression and Random Forest, while mental health issues and shared hobbies ranked higher in XGBoost. Trust score was the most important logistic predictor but ranked last in Random Forest. This instability suggests that the models were fitting noise rather than signal, and that no variable carried robust discriminative power in this dataset.

With the systematic literature review, nine thematic domains were identified as carrying genuine predictive signal that current datasets systematically omit: behavioural interaction markers, longitudinal trajectories of relationship satisfaction, physiological stress responses during conflict, adult attachment styles, intergenerational transmission of divorce, life events and stressors, sexual satisfaction, communication quality measured repeatedly over time, and validated psychometric instruments such as the Dyadic Adjustment Scale. Of these, the most consistently replicated and theoretically grounded are Gottman's Four Horsemen behavioural indicators, longitudinal relationship satisfaction trajectories, and attachment style, all measurable through validated instruments, yet largely absent from existing machine learning datasets.

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